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Lupus (2019) 28, 906–913
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Changing hospitalization trends for systemic lupus erythematosus and rheumatoid arthritis in England

J Hannah[1], J Galloway[2] and A Kaul[3]
[1]Department of Rheumatology, King's College Hospital, London, UK; [2]Department of Academic Rheumatology, King's College Hospital, London, UK; and [3]Department of Rheumatology, St George's University Hospital, London, UK

Objectives: Systemic lupus erythematosus (SLE) is a chronic, multisystemic, immune-mediated disorder associated with a substantial hospitalization risk. As a comparatively rare disease, there is a sparsity of research examining the burden of hospital admission in the contemporary era. We aim to describe national trends in hospitalization rates in England between 1998 and 2015 for SLE, using rheumatoid arthritis (RA) and general population rates as comparison cohorts for benchmarking. **Methods:** Hospital admission rates, emergency and day-case admission rates, length of stay and bed days used were calculated using finished consultant episodes from Hospital Episode Statistics data. Cochran-Armitage tests and linear regression quantified the significance and magnitude of change over time. **Results:** SLE admissions increased from 8.97 to 9.04 per 100,000 ($p < 0.001$) between 1998 and 2015. By comparison, RA admissions rose from 71.0 to 171.6 per 100,000 ($p < 0.001$) and all-cause admissions rose from 24,500 to 34,500 per 100,000 ($p < 0.001$). Emergency admissions decreased both for SLE (2.6 to 1.2 per 100,000) and RA (12.8 to 4.4 per 100,000) despite all-cause emergency admissions increasing from 9400 to 10,300 per 100,000. SLE and RA day cases increased, whilst median length of stay decreased. Despite increasing admissions, total bed days for SLE and RA fell by 60% and 90%, respectively. **Conclusions:** Whilst all-cause emergency admissions rose in the general population, those for SLE fell. Length of stay and bed days reduced and day cases increased, probably reflecting changing therapeutic strategies. This potentially large reduction in resource utilization warrants consideration when assessing the impact of new therapies. *Lupus* (2019) **28,** 906–913.

Key words: Admissions; health policy; hospitalization; rheumatoid arthritis; systemic lupus erythematosus

Significance and innovations

- Inpatient beds are an expensive resource and overall hospitalization rates are rising.
- However, emergency admissions and total bed days used in systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA) have significantly reduced.
- Advances in SLE and RA management have resulted in an increase in day case rates.

Correspondence to: Jennifer Hannah, King's College Hospital, Department of Rheumatology, Denmark Hill, London, SE5 9RS, UK.
Email: jennifer.hannah@doctors.org.uk
Received 28 March 2019; accepted 7 May 2019

Background

Systemic lupus erythematosus (SLE) is a complex, multisystemic autoimmune disorder responsible for considerable morbidity and has a profound impact on quality of life through reduced physical functioning, effects on mental and cognitive health, work productivity and an increased risk of hospitalization.[1,2]

Many SLE manifestations can lead to hospitalization, including nephropathy, neurological involvement, fever, serositis, or thrombosis.[3] Treatment regimens can also result in hospitalization, either as a mechanism for administering treatment (e.g. planned admission for parenteral biologic therapies) or due to side effects including infection and osteoporotic fractures.

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10.1177/0961203319853357

Patients with SLE or rheumatoid arthritis (RA) have a hazard ratio of 1.73 and 1.51 respectively for hospital admission compared with the general population.[1,4] For SLE admissions, the highest proportion of admissions are due to disease flare or infections, but surgery, childbirth and cardiovascular events are also common.[1,5] SLE patients have high readmission rates, with 16.5% of SLE patients being readmitted within 30 days,[6] although antimalarial use reduces these rates and length of stay.[7] Although there was improved survival of SLE patients over the latter half of the 20th century, this has plateaued and the subsequent effect on hospitalization rates is unclear.[8,9]

RA was chosen as a comparison for this study because, whereas it shares some pathological and clinical similarities to SLE, treatment algorithms and disease activity measures and their outcomes are more clearly defined than for SLE. RA most frequently presents as a symmetrical polyarthritis with synovial inflammation. Hospital admission may occur with severe articular inflammation or as a result of extraarticular manifestations such as pleuritis, pericarditis, vasculitis, scleritis or glomerulonephritis.[2,10] In RA higher rates of hospitalization are seen in those with signs of severe RA early in the disease process, such as seropositivity, erosions and nodules.[4] Unlike SLE, admission with disease flare is unusual because joint flares are often treated with oral or intramuscular corticosteroids. However, patients with RA are at higher risk of being admitted compared with the general population across a number of other categories including infections, endocrinological, haematological, respiratory, gastrointestinal, genitourinary, dermatological complaints and poisoning.[4] There is a particularly high risk for male RA patients to be admitted with depression (hazard ratio 7.16 compared with males without RA).[4]

The absolute number of all-cause emergency admissions in England rose 11.8% (1.35 million extra admissions) over the five-year period between 2004–2005 to 2008–2009, with population ageing estimated to account for only 40% of this increase.[11] Despite this, there are suggestions that admissions specifically for SLE and RA may be decreasing in other countries.[12–15] In Sardinia, between 2001 and 2012, there was a reduction in the number of admissions with a primary diagnosis of SLE, but an increase in cerebrovascular disease, acute coronary syndrome and chronic kidney disease admissions with comorbid SLE.[12] A similar nationwide study looking at RA patients in the United States estimated RA admission rates reduced from 13.9 to 4.6 per 100,000 between 1993 and 2011.[14] A single-centre study from New Zealand with patient-level data found a 75% reduction in admission rates for RA patients between 1990 and 1999, with a particular fall in admissions from acute RA flares or complications from 53% to 21% of total patients.[13] In contrast, total inpatient rheumatology workload appears to be increasing, with one United Kingdom (UK) tertiary hospital reporting 332 inpatient rheumatology referrals per annum in 2011 increasing month on month up to 502 in 2013, with RA and SLE being the second and fourth most common reason for new referrals, respectively.[16]

There has been significant progress in targeted treatments for RA but also SLE over the last 20 years. However, it is unclear whether these changes in clinical practice have had a recordable effect on hospitalization rates. It is possible that new biologic treatments may produce more toxicity and infections, whereas hospital admission may be required to administer some of the parenteral therapies.[17,18]

Factors which influence hospitalization rates include socioeconomic factors, background population health, funding and access to health care. In England, where health care is universally provided free-at-the-point-of access via the National Health Service (NHS), demand has been increasing on a wide range of related health services including outpatient attendances, elective inpatient admissions, primary care consultations and calls to ambulance services.[11]

Under the current global climate of increasing financial and political pressure on health care resources, the evolving usage of health care resources informs us about changing practice and its effect on the rheumatology workforce, allowing better planning of services in future. The purpose of this exploratory study was to describe national trends in hospitalization in a universally free-at-the-point-of access health care system over time, comparing planned and unplanned treatment episodes between SLE, RA and overall all-cause admissions.

Methods

This ecologic study uses longitudinal, population-based, observational data reported at summary level.

Admission data were extracted from Hospital Episode Statistics (HES), which reports all clinical activity in NHS hospitals, including private

patients treated in NHS hospitals and commissioned activity in the independent sector within England.[19] HES records an in-patient admission as a 'finished consultant episode' (FCE). This refers to a completed period of care within a single-provider institution, under a specific consultant, and terminates in either discharge, transfer or death.[19] An individual patient may have more than one period of care within the year. FCEs do not include emergency department attendances that did not result in admission.

These mandatory data are submitted centrally for annual analysis and publication. HES statistics are used for negotiating funding, allocation of resources, monitoring trends in activity and to produce performance and clinical indicators.[20]

A primary diagnosis International Statistical Classification of Diseases and Related Health Problems 10th Revision (ICD-10) code is recorded for each FCE. Secondary demographic, administrative and geographical data are available to describe episodes. This includes descriptors such as 'emergency', meaning unplanned admissions, 'day case', which refers to admissions which begin and end on the same day, 'mean' and 'median' length of stay (which exclude day cases), and 'bed days', which refers to a summation of the durations of all FCEs.[19]

Denominator population information for each year was obtained separately from the Office for National Statistics.[21]

The analysis included financial years 1998–2015 for all ICD-10 codes with subgroups of FCEs identified for primary diagnosis codes relating specifically to SLE and RA (Supplementary Table 1). The number of FCEs per denominator population provided the FCE rate.

The Cochran-Armitage test was used to assess for significant changes across time, and linear regression was used to describe the magnitude of change over time. The regression models were tested for validity using the Cook-Weisberg test for heteroscedasticity and the variance of residuals were plotted graphically. Where model diagnostics suggested nonlinear rates, spline models were used. To identify whether the change in FCE rate was being driven by a change in day case rate, a comparison of linear regression coefficients was performed using seemingly unrelated regression with a chi-squared test of significance.

Stata 15 was used for all analyses. Ethics approval was not required as per UK regulations as only publicly available data with no patient identifiable information were used.

Results

FCE rates

The mean age of SLE patients was younger than RA patients (40 vs 59 years).

SLE FCE rates increased from 8.97 in 1998/99 to 9.04 in 2014/15 per 100,000 (Figures 1(a) and (b)). There was a greater increase in RA FCE rates

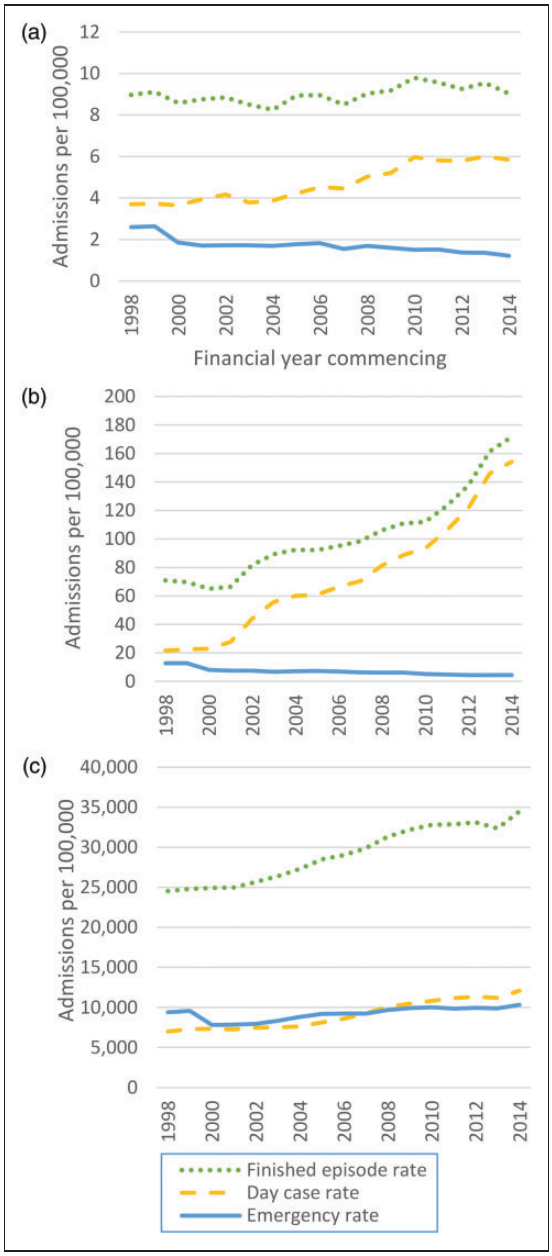

Figure 1 Finished consultant episode rates, day case rates and emergency rates according to primary diagnosis code: (a) Systemic lupus erythematosus. (b) Rheumatoid arthritis. (c) All diagnoses. All series show $p < 0.01$ using Cochran-Armitage test for trend.

from 70.96 to 171.63 per 100,000 although Cochran-Armitage test for trend shows both series are statistically significant positive trends ($p < 0.001$).

FCEs across all diagnoses also increased significantly from 12.0 million in 1998–1999 to 18.7 million in 2014–2015. Adjusting for population growth, this equates to 24,500 to 34,500 per 100,000 population (Figure 1(c)).

Linear regression showed that RA FCEs increased by 5.8 per 100,000 each year, and SLE by only 0.17. Model diagnostics suggested a nonlinear relationship. A cubic spline model demonstrated that the rate of RA episodes increased in a linear manner until 2010, but subsequently rose more rapidly (Supplementary Figure 1). Owing to the comparative rarity of SLE episodes, nonlinear models were not explored for these FCEs.

Day-case and emergency rates

Figure 1 compares the trends in emergency and day-case episodes. SLE day-case episodes increased by 58% between 1998 and 2015 (from 3.70 to 5.84 per 100,000). Meanwhile, SLE emergency episode rates decreased by 53% (from 2.61 to 1.22 per 100,000).

Similar trends are seen in RA emergency episodes, which reduced by 66% (from 12.82 to 4.39 per 100,000), whereas there was a substantial 610% increase in RA day cases (from 21.71 to 154.13 per 100,000). Comparing total RA FCEs with RA day-case episodes using seemingly unrelated regression shows the rise in day cases exceeds the rise in total RA FCEs ($p < 0.001$).

SLE and RA show a large reduction in emergency episode rates, contrary to national trends in emergency episodes which actually show a comparatively small 6% increase (from 9400 to 10,000 per 100,000). The Cochran-Armitage test for trend confirmed that the observed changes in FCE, day-case and emergency rates were statistically significant over time for all groups ($p < 0.001$).

Total bed days and length of stay

Over the 16-year study period, there was an annual average of 5.07×10^7 of bed days for all diagnoses. Only a very small proportion of these were attributable to SLE (13,500 days per annum) or RA (105,000 days per annum). The mean length of stay over the study period was 9.44, 8.84 and 8.46 days for all-diagnoses, SLE and RA, respectively (Figure 2). Figure 3 shows the trends in bed days. Between 1998 and 2015 there was a 60% reduction in total bed days for SLE and a 90% reduction in total bed days for RA, whereas all-diagnoses bed days decreased by only 5%.

Discussion

The total number of FCEs increased significantly for SLE, RA and all-causes. It is more pronounced in RA, with a 2.5-fold increase in admissions, than SLE. The increase in SLE and RA is driven by increasing day-case attendances, whereas emergency admissions in both categories fell. The overall consequence is that bed days for these conditions dropped sharply. In contrast, all-cause

Figure 2 Mean length of stay according to primary diagnosis code. RA: rheumatoid arthritis; SLE: systemic lupus erythematosus.

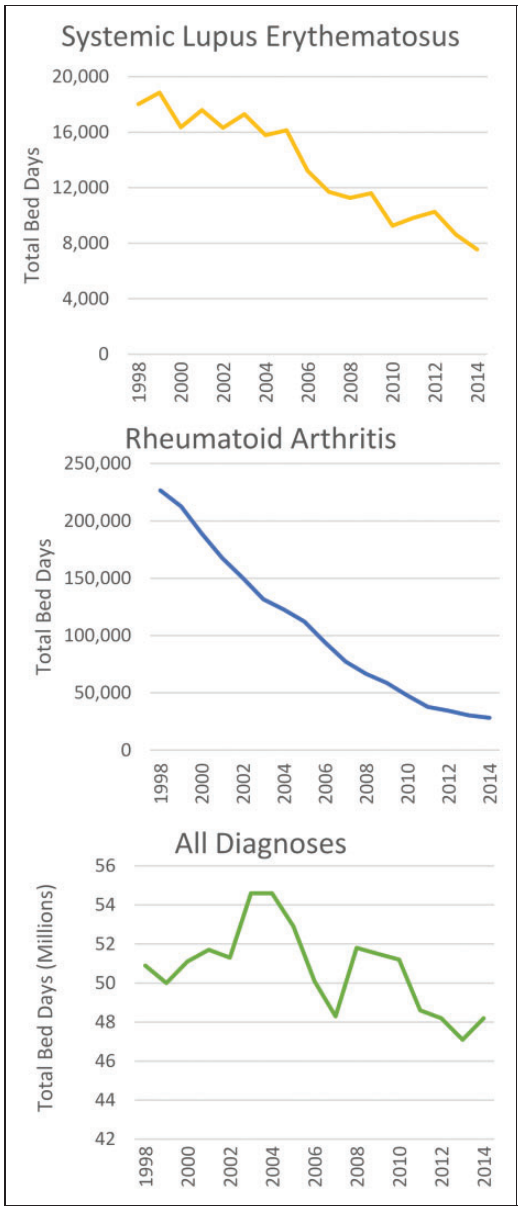

Figure 3 Total bed days used per year according to primary diagnosis.

emergency admissions increased with a small reduction in all-cause bed days.

There are likely to be several explanations for these results. SLE is a comparatively rare disease and the absolute number of episodes is low, so the admission trends are less dramatic than those seen in RA. The background incidence and prevalence of SLE may be increasing, but this is unlikely to sufficiently affect the trends seen.[22–24] Emergency admissions and bed days used are falling, which may suggest that management strategies have changed to prevent admissions. Whilst there is good evidence for improved clinical outcomes of several biologic agents in RA, the evidence is not so clear for SLE, and biologic use is not as widespread.[18] Figure 4 demonstrates that in the time-course of this study belimumab was the only newly licenced biologic for SLE (although rituximab is also used off-label) whereas several biologics including rituximab, infliximab, etanercept, adalimumab, abatacept, golimumab, tocilizumab and certolizumab pegol were used for RA.[25] In addition, older patients are more likely to require hospital admission.[11] The average age of RA patients in our study was older than for SLE. Therefore, population ageing may contribute more significantly to increasing FCEs in RA than SLE.

A New York study characterized 12.7% of SLE admissions as 'avoidable'. It also found that centres that saw a larger number of SLE patients had a smaller proportion of avoidable admissions.[26] In modern health care systems there is often a drive to focus care for complex SLE patients into specialist tertiary centres. This could contribute to a reduction in emergency admissions by providing more expert outpatient care and thereby reduce avoidable admissions. Other factors that can be addressed to reduce avoidable admissions include integration of primary and secondary care, improved clinician continuity, early senior clinician review in emergency departments and acute assessment units, integrating health and social care and encouraging patient self-management.[27] In practice, it is likely that a combination of the above factors is responsible for the observed effects.

In RA it is likely that changing therapeutic patterns are responsible for the observed trends. The background incidence of disease has changed little,[28] but biologic therapies have changed patient management strategies and the substantial increase in day-case episodes and reduction in bed days used probably reflects increased hospital attendance for parenterally administered therapy. This is supported by studies showing biologic prescriptions are rising. Tumour necrosis factor inhibitor prescribing increased 48-fold in Ireland between 2000 and 2010,[29] and total biologic prescriptions increased by 20% in an American study between 2004 to 2013.[30]

The falling emergency admission rates suggest that overall disease is being managed better in the community. Orthopaedic surgery rates are decreasing in the rheumatoid population, which also points towards better disease control with a reduction in accumulated joint damage.[29,31] Nonpharmacological management strategies, for example, promotion of early initiation of disease-modifying antirheumatic therapies and

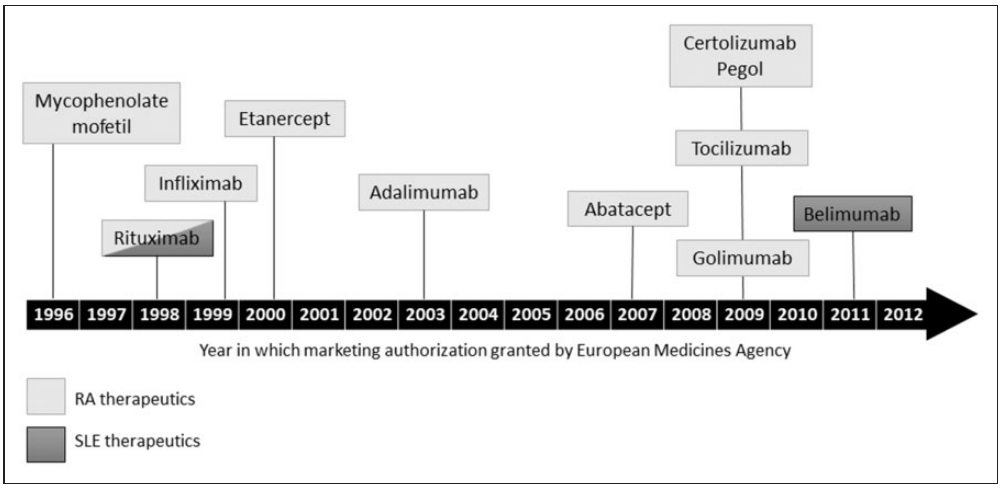

Figure 4 Newly introduced therapeutic agents for the treatment of systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA) over the time course of this study.

treat-to-target ethos to encourage tighter disease control, have played an important role in improving clinical outcomes.[32]

Reasons for increasing overall all-cause FCEs include an ageing population with increased frailty,[11] increased diagnostic and treatment possibilities,[33] and increased risk of litigation resulting in a lower clinical threshold for admission.[34] It has previously been reported that average length of stay is decreasing and fewer admissions are resulting in patient death, supporting the concept that admission threshold may be lowering.[11] Policies to improve efficiency and promote patient discharge cause an increase in available beds, which may create a paradoxical increase in admissions, a concept underpinned by 'Roemer's Law' in health care policy, that increasing the number of hospital beds per capita increases hospital utilization rates.[35] Financial cuts to social and primary care may explain the increasing number of patients self-referring to emergency departments. Systematic review has shown that increased primary care accessibility reduces avoidable hospital admissions for ambulatory care-sensitive conditions.[36] However, a randomized trial of complex patients with chronic conditions, likely more reflective of an RA and SLE cohort, showed that increased intensity primary care monitoring actually led to a paradoxical increase in readmissions over a six-month period.[37] Suggested reasons included increased detection of previously unnoticed medical problems or improved channels of communication between primary and secondary care.

Central policy changes to targets and payment systems in hospital, primary care and social care settings can change the case-mix of patients seen, as well as influence the way in which data are coded and reported. For example, in 2004 financial penalties were introduced in England if less than 98% of patients were seen within four hours in the emergency department. This was found to have an association with admission rates in some trusts, but not overall.[11]

Changes in coding quality are likely to have had some effect. In 2002 a government initiative called Payment-By-Results (PBR) introduced financial incentives to code data by paying hospitals according to amount of coded activity performed, potentially contributing to an apparent increase in activity. PBR improved coding accuracy from 73.8% to 96%; however, this does not fully explain our results as the observed trends predate 2002.[38] Steps have subsequently been taken to improve data quality, and it is now mandatory for health care providers to audit this annually. PBR may have encouraged the detailed recording of additional secondary diagnoses as complex patients with multiple comorbidities are reimbursed at a higher tariff rate. However, this will not affect primary diagnosis coding. Nationally, coding errors result in both under- and overpayments, suggesting there is no evidence of systemic financially motivated 'upcoding' errors.[39]

Limitations and strengths

No data were collected regarding secondary diagnoses and therefore we cannot comment on any changes to admission trends where SLE or RA are comorbidities. However, the advantage of

including only primary diagnosis ICD-10 codes is that all patients in this study had active and relevant disease and therefore trends are likely to relate to genuine disease and flare management strategies. Potential biases of poor-quality secondary diagnosis coding have therefore been avoided. External influences such as improvements to coding and data collection are immeasurable over time but should have been consistent across all categories. Our data cannot account for multiple episodes of the same patients as no patient-level data were available. This also makes the study susceptible to the ecologic fallacy and makes it hard to make any causal inference. However, the key strength of this study is the amount of data collected, which provides almost total coverage of the population of an entire country.

Conclusions

The increase in overall hospital admissions is stark and in keeping with other studies in many countries. However, the reduction in bed days, length of stay and emergency admission rates in SLE and RA is encouraging and implies that new treatments for these conditions are driving changing practice. Reductions in SLE admissions and bed days are clues that SLE care is evolving, but the changes lag behind those seen in RA, which likely reflects the boom in licencing of successful biologic therapies for RA and better evolved treat-to-target strategies.

Effective health care policy promoting improved outpatient care may have also contributed; however, these changes are not rheumatology specific, and similar reductions might be expected in all-cause emergency admissions, which have not been observed.

Owing to the tight financial constraints of a nationally funded health care system, future health care reforms need to address the increasing demands on services. Although there is significant variation in health care systems and local policies across the globe, services primarily publicly funded through taxation are common, with varying levels of privately funded contributions. Cost and efficiency savings are vital across all publicly funded services. Addressing preventable and avoidable admissions should be a priority. Inpatient bed-days are an expensive resource and are not frequently considered in a typical economic evaluation of new therapeutics. Reducing admissions could have a significant cost-saving impact, off-setting some of the cost of expensive biologic agents.

Future research needs to understand secondary causes of admission due to comorbidity or therapeutic side effects using patient-level data, as well as to explore the drivers of change and identify how management strategies can continue to evolve to serve patients as well as the health care economy.

Acknowledgement

This work was previously presented as an oral abstract at the British Society of Rheumatology Annual Conference, 2018.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

J Hannah https://orcid.org/0000-0002-3491-6653

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